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Measuring the Lifetime of Muon

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1. RESULTS AND DISCUSSION

Muon lifetime in measurement 1 is 2.09 ± 0.12 μs.
Muon lifetime in measurement 2 is 2.19 ± 0.13 μs.

This analysis uses 2 sets of data from the given data sheet, the first from row 14 of the data sheet and the second from row 12. There are 10 counts measurement in each data set. The number of counts follows the exponential decay formula. The number of count $N \propto e^{-t/\tau}$

$$\mathrm{N(t) = N_0 e^{-\frac{t}{\tau}}}$$

The number of counts was transformed into linear form to apply the least square fit, using the logarithms of the counts f(c) = ln(c) to fit in a linear function. The time of each count measurement is obtained by counting the number of bins. Each bin represents 0.5μs, and the midpoint of each bin was selected as the time of the corresponding counts(Bins$\times 0.5\mu s - 0.25\mu s$). For example, the time of the first counting point in $ln(c)_1$ was calculated by $1 \times 0.5\mu s - 0.25\mu s = 0.25\mu s$.

Table 1. Logarithm of counts obtained from given data

Time	$\mathrm{ln(c)_1}$	$\mathrm{ln(c)_2}$
μs	–	–
0.25	5.25 ± 0.07	5.19 ± 0.07
0.75	5.01 ± 0.08	4.93 ± 0.08
1.25	4.63 ± 0.10	4.79 ± 0.09
1.75	4.34 ± 0.11	4.45 ± 0.11
2.25	4.34 ± 0.11	4.28 ± 0.12
2.75	4.08 ± 0.13	3.85 ± 0.15
3.25	3.76 ± 0.15	3.91 ± 0.14
3.75	3.69 ± 0.16	3.85 ± 0.17
4.25	3.37 ± 0.19	3.47 ± 0.18
4.75	2.83 ± 0.24	3.04 ± 0.22

NOTE— $ln(c)_1$ comes from line 14 of data sheet, and $ln(c)_2$ comes from the line 12. $ln(c)$ is the logarithm of number of counts.

There are 10 counts in each set and the result after logarithm transformation are presented in Table 1. From Table 1, it can be seen that uncertainty increases with time in both $ln(c)_1$ and $ln(c)_2$

1.1. *Least Square Fit of data*

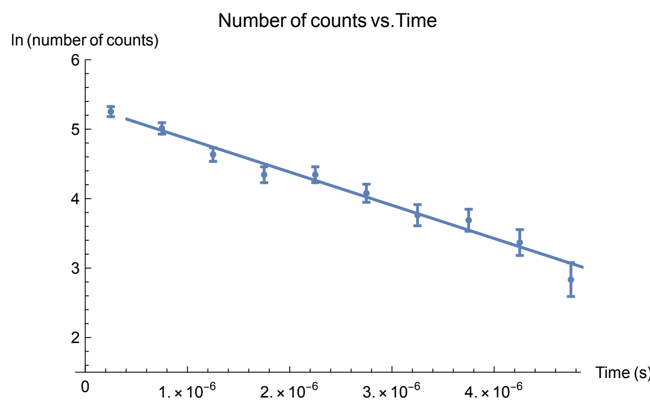

Figure 1. The best-fit slope m is -478117, with an uncertainty of 28220.5. The best-fit intercept b is 5.33825, with an uncertainty of 0.0574121.

The Figure 1 shows the linear relationship between the number of detected events and the counting time interval (after the log transformation of the number of counts). The lifetime of the muon is 2.09 ±0.12μs, obtained by calculating the negative of the reciprocal of the slope m $\tau = -\frac{1}{m}$.

The Figure 2 shows the linear relationship between the number of detected events and the counting time interval (after the log transformation of the number of counts). The lifetime of the muon is 2.19 ±0.13μs.

1.2. *Uncertainty*

The error bar shown on the plot indicates that the longer the detection time, the greater the uncertainty. This uncertainty may come from the PMT method counting the muon decay number. The timing clock is triggered when a muon goes from slowing down to stopping. A stopped muon will decay into an electron, a neutrino, and an antineutrino after a period of time.

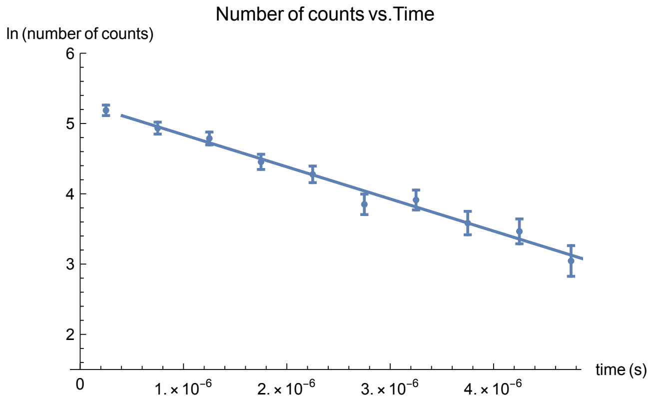

Figure 2. The best-fit slope is -456642, with an uncertainty of 28051.5. The best-fit intercept is 5.29679, with an uncertainty of 0.0580283.

The electron has the highest energy above the three, and its energy will trigger the second timing clock and generate a signal. The time interval between two triggers indicates the lifetime of a muon. For this reason, when the counting time increases, a larger amount of muons stop on the tube, while the counting start time varies, and the decay signals will overlap, causing uncertainty in the final data.

The original number of counts with the uncertainty is $number\, of\, counts \pm \sqrt{counts}$, which is $191 \pm \sqrt{191} = 191 \pm 14$ of the first data in the original line 14 for example.

In the analysis of Least Squre Fit, the uncertainty is propagated after the logarithm transformation. And the uncertainty propagation formula is given:

$$\sigma_f^2 = \sigma^2 x (\frac{\partial f}{\partial x})^2 + \sigma^2 y (\frac{\partial f}{\partial y})^2 + \sigma^2 z (\frac{\partial f}{\partial z})^2 ...$$

Therefore, the uncertainty of the logarithm transformation in y-axis is $\sigma_f = \sqrt{\sigma^2 x (\frac{\partial f}{\partial x})^2} = \sigma_x \cdot \frac{\partial f}{\partial x} = \sigma_x \cdot (ln(x))' = \sigma_x \cdot \frac{1}{x}$

The uncertainty of the 191 data point after the logarithm transformation $\frac{\sigma_x}{x}$ is $\frac{14}{191} = 0.07$. The complete expression of data point 191 with its uncertainty after the transformation is 5.25 $\pm$0.07.

In this experiment, N_0 and intercept b of the linear fit equation were not discussed.